

Abha Singh Sardar

M.Tech Smart Manufacturing
CGPA: 8.30



ACADEMIC PROFILE			
M.Tech - M.Tech Computer Science and Engineering	8.15	Maulana Azad National Institute of Technology	2017
M.Tech - M.Tech Smart Manufacturing	8.30	Indian Institute of Science	2027
B.Tech - Computer Science and Engineering	7.91	Kalinga Institute of Industrial Technology	2014
Class XII(ICSE)	75.67	Church School Beldih	2010
Class X(ICSE)	81.50	Church School Beldih	2008
COURSES			
Basics of Computing, AI and Data Science for Design and Manufacturing (PG)			
Sensors and Transducers (PG)			
Edge AI (PG)			
Introduction to Natural Language Processing (PG)			
PROJECTS			
Hybrid Encrypted Reversible Data Hiding (HERDH)		Research Scholar	Aug'16-Jul'17
Master's Thesis project: Developed a hybrid reversible data-hiding method using wavelet-domain transformation, achieving >50 dB PSNR and a 72% improvement in visual fidelity over existing techniques.			
Hybrid CNN-ViT Model for Plant Disease Classification		Research Scholar	May'23-Aug'23
- Developed a <i>hybrid architecture integrating Convolutional Neural Networks for local feature extraction with Vision Transformers for long-range dependency modelling</i>, achieving a 98.23% test accuracy on the Tomato Leaf Disease Classification dataset containing 21,000 images and Plant Disease Recognition. - Reduced test loss significantly on the <i>Plant Disease Recognition (0.3182 to 0.0956)</i> and <i>Tomato Leaf Disease (0.2108 to 0.0343)</i> datasets, while improving F1 scores from 0.8842 to 0.9819 and 0.9608 to 0.9947, respectively, compared to baseline CNN models.			
Recommendation System for E-Commerce using Truncated SVD-SGD			Jan'26-Jan'26
- Developed a model-based recommendation engine using Truncated Singular Value Decomposition (SVD) to decompose a sparse utility matrix of 50,000 user-product interactions into latent feature vectors. - Implemented item-to-item correlation analysis to generate targeted product suggestions, identifying highly relevant items with a Pearson correlation coefficient exceeding 0.90 across a catalog of 2,845 beauty products.			
IMU-Based Human Activity Recognition (HAR) System on Edge			Jan'26-Feb'26
- Developed an end-to-end TinyML pipeline on Arduino Nicla Vision to classify 12 distinct human activities, such as eating and pick-and-place. A dataset of over 120,000 IMU samples was collected at 50 Hz using a 3-axis Accelerometer and 3-axis Gyroscope. - Performed statistical feature extraction (e.g., Mean, RMS, Energy) on raw data, aggregated into a 72-dimensional vector over 20-sample sliding windows to train a Decision Tree classifier, and deployed real-time on-device inference with 99.43% classification accuracy for high-dynamic activities such as jogging and cycling.			
CERTIFICATIONS			
Google Data Analytics Professional Certificate		Coursera	2024
DeepLearning.AI TensorFlow Developer		Coursera	2022
Machine Learning		Coursera	2022
Python for Data Science, AI & Development		Coursera	2023
ADDITIONAL SECTION			
- Served as a Meta Reviewer for the IEEE International Students' Conference SCEECS'25 held on January 18th-19th, 2025, recognized for exceptional service in upholding research standards. Sardar, A. S., & Ranjan, V. (2023, November). Enhancing computer vision performance: A hybrid deep learning approach with CNNs and vision transformers. In International Conference on Computer Vision and Image Processing (pp. 591-602). Cham: Springer Nature Switzerland. - Tiwari, N., & Sardar, A. S. (2017). Advancement in Reversible Data Hiding techniques. International journal of Computer Science and Network Security, 17(3). - Ranjan, A., Sardar, A. S., Ranjan, V., & Sharma, N. (2025, November). Improving phishing detection using attention-augmented CNN-GRU-RNN networks. In 5th International Conference on Communication, Computing and Electronics Systems (ICCCES 2026) (Accepted on 13 November 2025). - Served as Moderator for the Ripples'26 Panel Discussion at IISc Open Day on "Beyond Automation: Artificial Intelligence in Engineering Design and Production. Skills: Python, Data Analytics, TinyML, Model Optimization and Deployment, TensorFlow, Research and Scientific Writing, Machine Learning and Deep Learning, Computer Vision.			
LANGUAGES		English, Hindi	